Yongjin Yang

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Research Focus

I study RL post-training for LLMs and self-improving agents, focusing on curricula and exploration in multi-domain RLVR, knowledge distillation, prompt optimization, and memory-augmented agents.

Education

University of Toronto — Toronto, Canada
Ph.D. in Computer Science | Advisor: Zhijing Jin | GPA: 4.0/4.0 — *Sep 2025–Summer 2029 (expected)*

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
M.S. in Artificial Intelligence | Advisors: Se-Young Yun and Kimin Lee — *Mar 2023–Feb 2025*
GPA: 4.18/4.3 | Ranked 2nd in the department

Seoul National University — Seoul, South Korea
B.S. in Electrical and Computer Engineering | GPA: 4.12/4.3 | *Summa Cum Laude* — *Mar 2017–Feb 2023*

Research Experience

ServiceNow AI Research — Montreal, Canada
Visiting Researcher, UI Assist Team, supervised by Alexandre Drouin and Christopher Pal — *Feb 2026–Sep 2026*

- Led **APT**, an RL framework that uses evolving prompts as training-time teachers and distills their guidance into the weights. On HoVer, WebShop, and SATBench with Qwen3-8B and Gemma3-12B, it improved over GRPO by 6.5 points on average and outperformed the strongest baseline in all six settings.
- Developing a self-improvement loop for software-engineering agents on R2E-Gym, driven by meta-exploration (ongoing).

University of Toronto & Vector Institute — Toronto, Canada
Graduate Researcher, advised by Zhijing Jin — *Sep 2025–Present*

- Developed **TAC**, an adaptive curriculum that selects RL training domains based on cross-domain transfer. Improved average accuracy by up to 2.8 points over a learnability-only baseline on Qwen3-1.7B and Llama-3.2-3B, with less than 1% additional training time (NeurIPS 2026).
- Built **SFD**, a hybrid LLM-to-SLM distillation method that uses the student's own uncertainty (token entropy and NLL) to decide where to imitate teacher solutions and where to rewrite them in its own words. It outperformed off-policy distillation and on-policy rejection fine-tuning on six reasoning benchmarks.
- Co-authored collaborative projects on long-horizon reasoning with skill-aware RL (Skill-Native LLMs, NeurIPS 2026), dense supervision through self-revision (Self-Distillation Zero, ICML 2026 RLxF Workshop Oral), and multi-agent LLM behavior (Agent-to-Agent Theory of Mind, EMNLP 2025; Corrupted by Reasoning, COLM 2025).

Naver AI Lab — Seongnam, South Korea
Research Intern, supervised by Hwaran Lee — *Mar 2024–Jun 2024*

- Led **MAQA**, a multi-answer benchmark covering world knowledge, math, and commonsense reasoning. Used it to compare five uncertainty-quantification methods for white- and black-box LLMs under data uncertainty (NAACL 2025 Findings).
- Co-developed **CSRT**, a multilingual red-teaming framework tested on 10 LLMs with code-switching prompts in up to 10 languages. It produced 46.7% more successful attacks than English-only red teaming (ACL 2025 Main).

KAIST Seoul, South Korea
Graduate Researcher, advised by Se-Young Yun and Kimin Lee *Mar 2023–Feb 2025*

- Designed and built **EXIF**, an exploration-first loop that turns environment interactions into training data and improves agent skills through iterative feedback. Led the main WebShop and Crafter experiments and analysis (co-first author).
- Designed and built **FiFA**, which filters preference data by reward margin, text quality, and diversity before diffusion-model alignment. Led the main experiments and analysis as co-first author. In human evaluation, FiFA's win rate exceeded full-data training by 17 percentage points while using less than 0.5% of the data and 1% of the GPU hours (ICLR 2025).
- Developed transfer-learning methods for vision-language models: **APEX** (EMNLP 2024) and normalization-aware adapters (AAAI 2024).

Seoul National University Seoul, South Korea
Undergraduate Researcher, advised by Taesup Kim *Jun 2022–Sep 2022*

- Developed **MELO**, adaptive loss reweighting for imbalanced cold-start recommendation (CIKM 2023 Oral, co-first author).

Publications

*Equal contribution †Corresponding author C: Conference P: Preprint

Conference Papers

[C14] **Transferability for General Reasoning: An Automated Curriculum for Multi-Domain RLVR.**
Yongjin Yang, Jiarui Liu, Yinghui He, Lechen Zhang, Bernhard Schölkopf, Zhijing Jin†.
NeurIPS 2026. Also ICML 2026 RLxF Workshop, *Best, 5 of 222 submissions*. [paper] [code]

[C13] **Toward Skill-Native LLMs: Skill Entropy for Benchmarking and Training Long-Horizon Reasoning.**
Yinghui He, Ling Yang, Jiarui Liu, **Yongjin Yang**, et al.
NeurIPS 2026. [paper] [code]

[C12] **LLM-as-a-Tutor: Policy-Aware Prompt Adaptation for Non-Verifiable RL.**
Yujin Kim*, Namgyu Ho*, Sangmin Hwang*, Joonkee Kim, **Yongjin Yang**, Sangmin Bae, Seungone Kim, Jaehun Jung, Se-Young Yun†, Hwanjun Song†.
NeurIPS 2026. [paper]

[C11] **Entropy-Aware On-Policy Distillation of Language Models.**
Woogyeol Jin, Taywon Min, **Yongjin Yang**, Dennis Wei, Yi Zhou, Swanand Ravindra Kadhe, Nathalie Baracaldo, Kimin Lee.
ICML 2026. [paper] [code]

[C10] **Agent-to-Agent Theory of Mind: Testing Interlocutor Awareness among Large Language Models.**
Younwoo Choi*, Changling Li*, **Yongjin Yang**, Zhijing Jin.
EMNLP 2025 Main. [paper] [code]

[C9] **Corrupted by Reasoning: Reasoning Language Models Become Free-Riders in Public Goods Games.**
David Guzman Piedrahita, **Yongjin Yang**, Mrinmaya Sachan, Giorgia Ramponi, Bernhard Schölkopf, Zhijing Jin.
COLM 2025. [paper] [code]

[C8] **Self-Training Elicits Concise Reasoning in Large Language Models.**
Tergel Munkhbat*, Namgyu Ho*, Seo Hyun Kim*, **Yongjin Yang**, Yujin Kim, Se-Young Yun.
ACL 2025 Findings. [paper] [code]

[C7] **Code-Switching Red-Teaming: LLM Evaluation for Safety and Multilingual Understanding.**
Haneul Yoo, **Yongjin Yang**, Hwaran Lee.
ACL 2025 Main. [paper] [code]

[C6] **Automated Filtering of Human Feedback Data for Aligning Text-to-Image Diffusion Models.**
Yongjin Yang*, Sihyeon Kim*, Hojung Jung, Sangmin Bae, SangMook Kim, Se-Young Yun†, Kimin Lee†.
ICLR 2025. [paper] [code]

[C5] **MAQA: Evaluating Uncertainty Quantification in LLMs Regarding Data Uncertainty.**
Yongjin Yang, Haneul Yoo, Hwaran Lee.
NAACL 2025 Findings. [paper] [code]

[C4] **Towards Difficulty-Agnostic Efficient Transfer Learning for Vision-Language Models.**
Yongjin Yang*, Jongwoo Ko*, Se-Young Yun.
EMNLP 2024 Main. [paper] [code]

[C3] **Leveraging Normalization Layer in Adapters with Progressive Learning and Adaptive Distillation for Cross-Domain Few-Shot Learning.**
Yongjin Yang, Taehyeon Kim, Se-Young Yun.
AAAI 2024. [paper]

[C2] **HARE: Explainable Hate Speech Detection with Step-by-Step Reasoning.**
Yongjin Yang*, Joonkee Kim*, Yujin Kim*, Namgyu Ho, James Thorne†, Se-Young Yun†.
EMNLP 2023 Findings. [paper] [code]

[C1] **Meta-Learning with Adaptive Weighted Loss for Imbalanced Cold-Start Recommendation.**
Minchang Kim*, **Yongjin Yang***, Jung Hyun Ryu, Taesup Kim.
CIKM 2023. *Oral*. [paper] [code]

Preprints

[P5] **Learning from Evolving Prompts: Adaptive Prompt Teaching for Reinforcement Learning.**
Yongjin Yang, Lechen Zhang, Jiarui Liu, Yinghui He, Christopher Pal, Alexandre Drouin, Zhijing Jin.
Under review.

[P4] **Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision.**
Yinghui He, Simran Kaur, Adithya Bhaskar, **Yongjin Yang**, Jiarui Liu, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, Sanjeev Arora.
Preprint. ICML 2026 RLxF Workshop, *Best, 5 of 222 submissions*. [paper] [code]

[P3] **Making Complex Reasoning Student-Friendly: A Hybrid Distillation Framework for Small Language Models.**
Yongjin Yang, Yinghui He, Jiarui Liu, Zhijing Jin.
Preprint. ICLR 2026 Workshop on Scaling Post-Training for LLMs. [paper]

[P2] **Automated Skill Discovery for Language Agents through Exploration and Iterative Feedback.**
Yongjin Yang*, Sinjae Kang*, Juyong Lee, Dongjun Lee, Se-Young Yun†, Kimin Lee†.
Preprint. [paper]

[P1] **Revisiting Multi-Agent Debate as Test-Time Scaling: A Systematic Study of Conditional Effectiveness.**
Yongjin Yang*, Euiin Yi*, Jongwoo Ko, Kimin Lee†, Zhijing Jin†, Se-Young Yun†.
Preprint. ICML 2025 Multi-Agent Systems Workshop. [paper] [code]

Selected Talks

- "Toward Skill-Native LLMs: Skill Entropy for Benchmarking and Training Long-Horizon Reasoning." Papers in the Park, Cohere Labs Open Science Community, Aug 2026.
- "Toward General Reasoning Across Domains." Invited talk, LawZero, Aug 2026.
- "Making Complex Reasoning Student-Friendly." Lightning talk, Remarkable 2026, Vector Institute, Feb 2026.
- "The Limitations of RL for LLMs in Achieving AI for Science." Invited talk, Trajectory Lab, Nov 2025. [video]
- "Capability and Safety of LLMs: Bottlenecks of RL and Multi-Agent Simulacra." Research Connections (with Max Kleiman-Weiner), Cohere Labs Open Science Community, Sep 2025.

Teaching

Teaching Assistant, Probabilistic Learning and Reasoning (CSC412), University of Toronto *Sep 2026–Dec 2026*
Teaching Assistant, Introduction to Programming Languages, University of Toronto *Sep 2025–Dec 2025*
Departmental Tutor, Introduction to Algorithms, Seoul National University *Sep 2021–Feb 2022*
Group Tutor, Principles of Physics I, Seoul National University *Mar 2018–Aug 2018*

Honors & Awards

Connaught International Scholarship, University of Toronto *2025*
ICML Gold Reviewer *2026*
Korea National Science and Technology Scholarship, full-ride academic scholarship *2017–2023*

Service

Reviewer: NeurIPS, ICML, ICLR, COLM, TMLR, and ACL Rolling Review *2024–2026*
Lab Administrator, Jinesis Lab, University of Toronto: manages the lab's HPC compute allocations and infrastructure

Technical Skills

Research: LLM post-training, RLVR, exploration, multi-task RL, knowledge distillation, prompt optimization, preference learning, language agents
Programming/ML: Python, C++, PyTorch, Hugging Face Transformers/Diffusers, DeepSpeed, Megatron-LM, veRL, vLLM, Ray
Systems: CUDA, SLURM, Docker, Kubernetes, Git